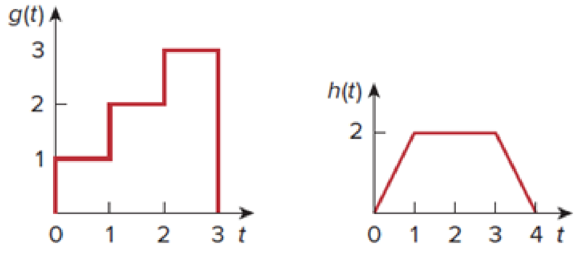
Chapter 15, Problem 18P

Problem

Obtain the Laplace transforms of the functions in Fig. 15.30.

Figure 15.30:



(a) (b)

Step-by-step solution

Step 1 of 4

(a) The following is the given function:

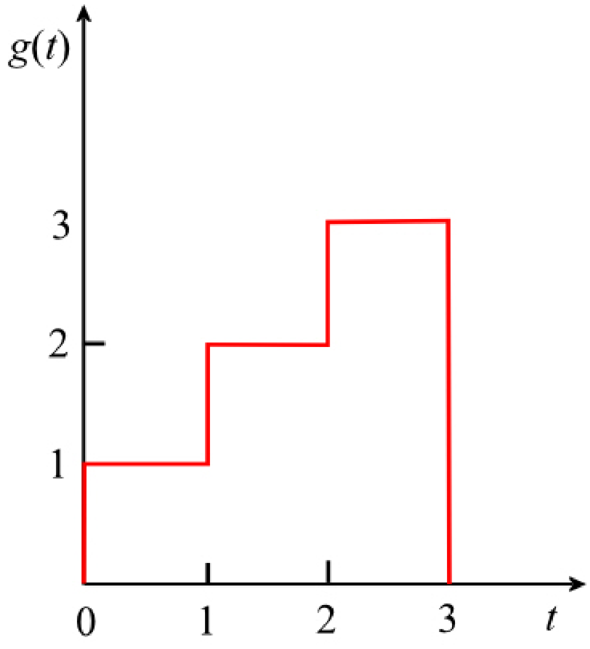


Figure 1

Step 2 of 4

The given function $g(t)$ can be expressed as,

$$g(t)=\begin{cases}1 & \text{for } 0\le t\le 1\\ 2 & \text{for } 1\le t\le 2\\ 3 & \text{for } 2\le t\le 3\end{cases}$$

The Laplace transform of $g(t)$ is

$$\begin{aligned}G(s)&=\int_0^\infty g(t)e^{-st}dt\\ &=\int_0^1(1)e^{-st}dt+\int_1^2(2)e^{-st}dt+\int_2^3(3)e^{-st}dt\\ &=\int_0^1 e^{-st}dt+2\int_1^2 e^{-st}dt+3\int_2^3 e^{-st}dt\\ &=\left.\frac{e^{-st}}{-s}\right|_0^1+2\left(\left.\frac{e^{-st}}{-s}\right|_1^2\right)+3\left(\left.\frac{e^{-st}}{-s}\right|_2^3\right)\end{aligned}$$

$$=\frac{-1}{s}\left[e^{-s}-e^{0}+2\left(e^{-2s}-e^{-s}\right)+3\left(e^{-3s}-e^{-2s}\right)\right]$$

$$=\frac{-1}{s}\left[e^{-s}-1+2e^{-2s}-2e^{-s}+3e^{-3s}-3e^{-2s}\right]$$

$$=\frac{-1}{s}\left[-e^{-s}-e^{-2s}+3e^{-3s}-1\right]$$

$$=\frac{1}{s}\left[1+e^{-s}+e^{-2s}-3e^{-3s}\right]$$

Thus the Laplace transform of $g(t)$ is $\boxed{\frac{1}{s}\left[1+e^{-s}+e^{-2s}-3e^{-3s}\right]}$.

Step 3 of 4

(b) The following is the given function:

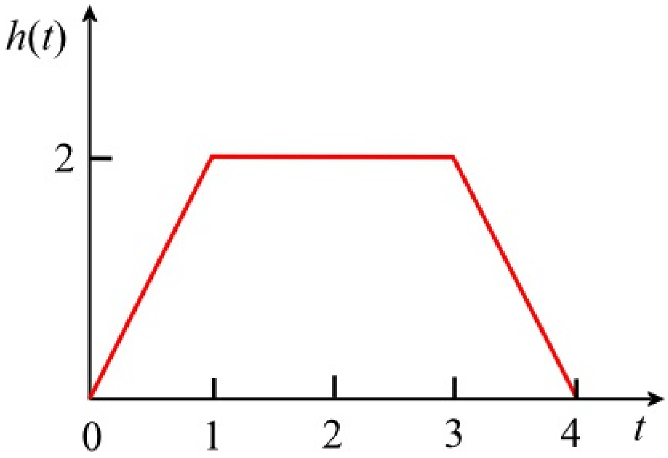


Figure 1

Step 4 of 4

The given function $h(t)$ can be expressed as,

$$h(t)=\begin{cases}2t & \text{for } 0\le t\le 1\\ 2 & \text{for } 1\le t\le 3\\ -2t+8 & \text{for } 3\le t\le 4\end{cases}$$

The Laplace transform of $h(t)$ is

$$\begin{aligned}H(s)&=\int_0^\infty h(t)e^{-st}dt\\
&=\int_0^1(2t)e^{-st}dt+\int_1^3(2)e^{-st}dt+\int_3^4(-2t+8)e^{-st}dt\\
&=2\int_0^1 te^{-st}dt+2\int_1^3 e^{-st}dt-2\int_3^4 te^{-st}dt+8\int_3^4 e^{-st}dt\\
&=\left\{\begin{array}{l}2\left[t\dfrac{e^{-st}}{-s}\Big|_0^1-\displaystyle\int_0^1\dfrac{e^{-st}}{-s}(1)\,dt\right]+2\left(\dfrac{e^{-st}}{-s}\Big|_1^3\right)-\\ 2\left[t\dfrac{e^{-st}}{-s}\Big|_3^4-\displaystyle\int_3^4\dfrac{e^{-st}}{-s}(1)\,dt\right]+8\left(\dfrac{e^{-st}}{-s}\Big|_3^4\right)\end{array}\right\}\end{aligned}$$

Simplify further,

$$\begin{aligned}H(s)&=\left\{\begin{array}{l}2\left[\dfrac{e^{-s}}{-s}-\left(\dfrac{e^{-st}}{s^2}\Big|_0^1\right)\right]+2\left(\dfrac{e^{-3s}-e^{-s}}{-s}\right)-\\ 2\left[\dfrac{4e^{-4s}-3e^{-3s}}{-s}-\left(\dfrac{e^{-st}}{s^2}\Big|_3^4\right)\right]+8\left(\dfrac{e^{-4s}-e^{-3t}}{-s}\right)\end{array}\right\}\\
&=\left\{\begin{array}{l}2\left[\dfrac{e^{-s}}{-s}-\left(\dfrac{e^{-s}-1}{s^2}\right)\right]+\dfrac{2e^{-3s}-2e^{-s}}{-s}-\\ 2\left[\dfrac{4e^{-4s}-3e^{-3s}}{-s}-\left(\dfrac{e^{-4s}-e^{-3s}}{s^2}\right)\right]+\dfrac{8e^{-4s}-8e^{-3s}}{-s}\end{array}\right\}\\
&=\left\{\begin{array}{l}\dfrac{2e^{-s}}{-s}-\left(\dfrac{2e^{-s}-2}{s^2}\right)+\dfrac{2e^{-3s}-2e^{-s}}{-s}-\\ \dfrac{8e^{-4s}-6e^{-3s}}{-s}+\left(\dfrac{2e^{-4s}-2e^{-3s}}{s^2}\right)+\dfrac{8e^{-4s}-8e^{-3s}}{-s}\end{array}\right\}\\
&=\left\{\begin{array}{l}\dfrac{2e^{-s}+2e^{-3s}-2e^{-s}-8e^{-4s}+6e^{-3s}+8e^{-4s}-8e^{-3s}}{-s}-\\ \dfrac{2e^{-s}-2-2e^{-4s}+2e^{-3s}}{s^2}\end{array}\right\}\\
&=0-\frac{-2+2e^{-s}+2e^{-3s}-2e^{-4s}}{s^2}\\
&=\frac{2}{s^2}\left(1-e^{-s}-e^{-3s}+e^{-4s}\right)\end{aligned}$$

Thus the Laplace transform of $h(t)$ is $\boxed{\dfrac{2}{s^2}\left(1-e^{-s}-e^{-3s}+e^{-4s}\right)}$.